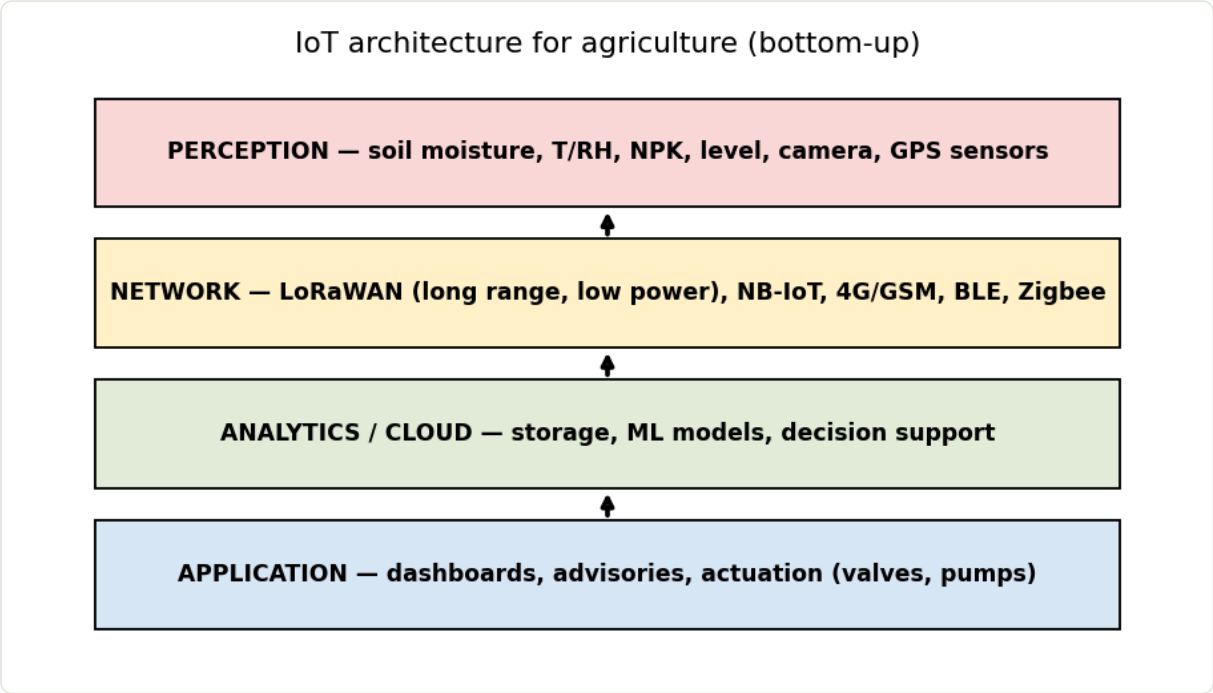
UNIT 2 · TOPIC 10(C)

# IoT (Internet of Things) in Agriculture

*Syllabus topic: Machine learning, IoT, robotics and mechatronics in farm machinery*

**IoT** = networked physical devices with sensors + connectivity that collect and exchange data for monitoring and automatic control, usually through a 4-layer architecture.




*IoT architecture for agriculture (bottom-up)*

## Connectivity choices

| Technology | Use |
|---|---|
| LoRaWAN | Kilometres of range at milliwatt power — ideal for field sensor networks |
| NB-IoT / 4G | Cellular coverage |
| BLE / Zigbee | Short range |
| Satellite | Remote telemetry |

**Edge computing** processes data on or near the device to cut latency and data cost.

## FMP applications

- Sensor-based smart irrigation (soil-moisture-triggered valves).
- Machinery **telematics** (location, hours, fuel, fault codes of tractor fleets).
- Cold-chain and silo monitoring.
- Livestock wearables.
- Weather stations feeding spray-decision tools.

## Practice MCQs

**Q199. The IoT protocol designed for kilometres of range at very low power is:**

(a) Bluetooth
(b) LoRaWAN
(c) USB
(d) NFC

**Q203. Edge computing in IoT means processing data:**

(a) Only in cloud
(b) On or near the sensing device
(c) On paper
(d) After harvest

**Answers**

Q199 — (b) LoRaWAN

Q203 — (b) On or near the sensing device

**Sources:** Kepner et al. (hydraulic principles); Drone Rules 2021 (Ministry of Civil Aviation); MoA&FW SOPs for drone application of pesticides & nutrients; SMAM drone guidelines & PIB releases (Kisan Drone, Namo Drone Didi — verified July 2026); ISO 11783 (ISOBUS); NPTEL lectures on mechatronics & precision agriculture.